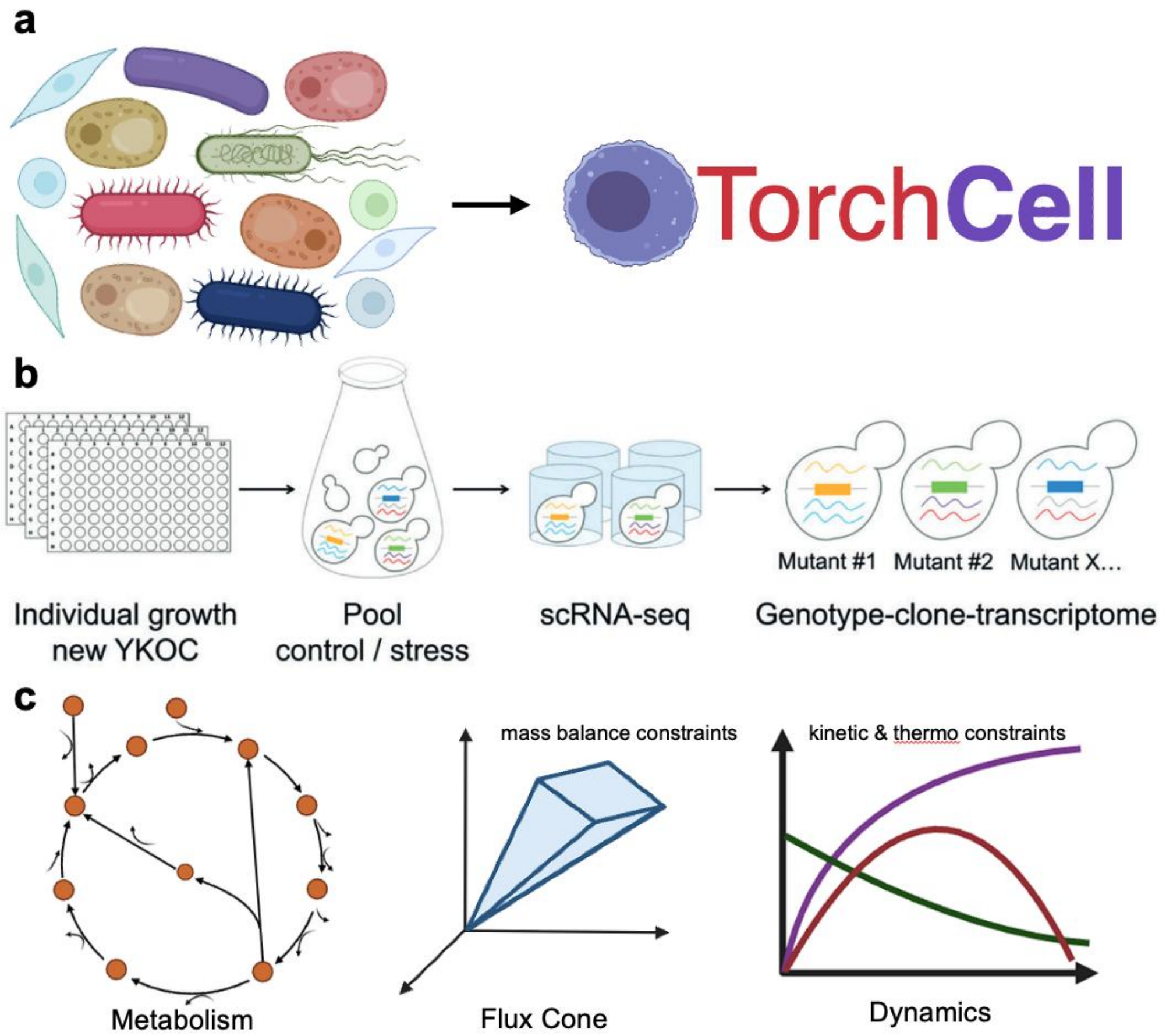

a
TorchCell
b
Mutant #1
Mutant #2
Mutant X…
Individual growth
new YKOC
Pool
control / stress
scRNA-seq
Genotype-clone-transcriptome
c
Metabolism
mass balance constraints
Flux Cone
kinetic & thermo constraints
Dynamics